Business-Insight Report: Customer Churn Prediction and Retention

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Dataset: IBM Telco Customer Churn (7,043 customers, 21 columns)

Live app: https://fp-churn-tsqxrn6e86avqpcszde6rc.streamlit.app

GitHub: https://github.com/zaidali21082007-spec/fp-churn

1. Summary

About 26.54% of customers (1,869 of 7,043) churned, representing $139,130.85 of monthly revenue. Churn is concentrated in month-to-month contracts and among customers who pay manually rather than by automatic payment. A Random Forest model reaches a ROC-AUC of 0.844 on the held-out test set. On that 1,409-customer test set it flags 296 High or Critical risk customers carrying about $15,030 of expected monthly revenue at risk. The Retention Dashboard scores all 7,043 customers, including the training data, and flags 1,477 High or Critical customers with $76,429.55 of expected monthly revenue at risk; those larger figures are in-sample. The recommendations below target the segments where churn is concentrated and where the data supports action.

2. Key Findings

Measure	Result
Overall churn rate	26.54% (1,869 of 7,043 customers)
Churned monthly revenue	$139,130.85 (of $456,116.60 total monthly revenue)
Churn rate: month-to-month / one-year / two-year	42.71% / 11.27% / 2.83%
Month-to-month share of churned monthly revenue	$120,847 of $139,131 (86.9%)
Churn rate: automatic payment vs manual payment	16.0% vs 34.7%
Auto-pay gap: month-to-month / one-year / two-year	13.0 / 2.7 / 0.1 percentage points
Churn rate by tenure group: New / Developing / Established / Loyal	47.44% / 28.71% / 20.39% / 9.51%
Churn rate by segment: Senior Solo / Senior Family / Solo / Family	49.2% / 34.42% / 31.15% / 17.16%
Internet customers without vs with Tech Support	41.6% vs 15.2%

3. Model Performance

Four classifiers were trained on the same 80/20 stratified split (test set: 1,409 customers, 374 churned and 1,035 retained). Random Forest was selected by ROC-AUC, with recall as the tie-breaker. The four models are close on ROC-AUC, so the choice is a narrow one.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest (selected)	**0.804**	**0.665**	**0.527**	**0.588**	**0.844**
Logistic Regression	0.797	0.646	0.521	0.577	0.843
Decision Tree	0.797	0.668	0.468	0.550	0.838
K-Nearest Neighbors	0.777	0.582	0.570	0.576	0.821

Accuracy is not the basis for selection, because only 26.54% of customers churn. At the default 0.50 threshold, the Random Forest catches 197 of 374 churners (recall 52.67%), and 66.55% of the customers it flags do churn (precision). Recall matters most here: a missed churner is a lost customer, while a false alarm only costs one retention contact. Lowering the threshold to 0.30 raises recall to 77.81% (291 of 374 churners) while flagging 543 customers at 53.59% precision. The Threshold Analysis page lets the team choose the cutoff that fits its outreach capacity.

On the held-out test set, actual churn rises steadily across the risk tiers:

Risk tier	Customers	Actual churn rate	Share of test set
Low	809	8.2%	57.4%
Medium	304	36.5%	21.6%
High	229	62.9%	16.3%
Critical	67	79.1%	4.8%

4. Retention Recommendations

Recommendation 1: Offer annual-contract discounts to month-to-month customers

Finding: Month-to-month customers churn at 42.71%, about 15 times the two-year rate of 2.83%.

Evidence: Churn Dashboard, revenue at risk by contract table; Churn Analysis, churn rate by contract chart.

Recommendation: Offer a discounted one-year or two-year contract, starting with High and Critical risk month-to-month customers.

Expected Result: Month-to-month customers account for $120,847 of the $139,131 churned monthly revenue. As an illustration, retaining 20% of that amount would be worth about $24,169 a month.

Recommendation 2: Move manual-payment customers to automatic payment

Finding: Automatic payment is associated with lower churn: 16.0% versus 34.7% for manual payers. The gap is 13.0 points on month-to-month contracts, 2.7 on one-year and 0.1 on two-year contracts, so contract type explains much of it.

Evidence: Churn Analysis, auto-pay chart; within-contract gap table in the notebook, Section 4.1.

Recommendation: Offer a one-time bill credit to manual-payment customers who switch to automatic payment. Keep this offer separate from the contract discount so each effect can be measured.

Expected Result: Expect a smaller benefit than the raw 18.7-point difference suggests. Confirm the effect with a pilot group before scaling.

Recommendation 3: Prioritize Senior Solo customers for outreach

Finding: Senior Solo customers have the highest churn rate (49.2%) and the highest share flagged High or Critical risk (50.45%). Solo customers account for the largest flagged count (723).

Evidence: Churn Analysis, churn rate by customer segment; Retention Dashboard, high-risk customers by segment table (in-sample scoring, all 7,043 customers).

Recommendation: Assign a named account contact or proactive check-in call to Senior Solo customers in the High and Critical tiers.

Expected Result: Retention effort goes to the segment with the highest concentration of risk. Measure whether churn among contacted customers falls relative to a control group.

Recommendation 4: Use the risk tiers to set outreach order

Finding: On held-out data, actual churn is 8.2% in the Low tier and 79.1% in the Critical tier. Among High and Critical customers, predicted churn (65.0%) is close to observed churn (66.55%).

Evidence: Retention Dashboard, risk tiers; Model Comparison, ROC-AUC 0.844; calibration table in the notebook, Section 6.

Recommendation: Contact Critical customers first, then High. Use the Threshold Analysis page to choose a cutoff that matches the team's capacity.

Expected Result: Retention effort goes to the customers most likely to leave, rather than being spread across the whole base.

Recommendation 5: Run the monthly high-risk campaign with a control group

Finding: About 21% of customers fall in the High or Critical tiers. At the default 0.50 threshold, the model flags 296 of the 1,409 test-set customers.

Evidence: Retention Dashboard, downloadable high-risk customer list; Threshold Analysis, precision and recall at each threshold.

Recommendation: Export the high-risk list each month, contact a sample of it, and hold back a control group that receives no outreach.

Expected Result: The business learns which action works and can measure the revenue it protects, rather than relying on the model score alone.

5. Limitations

- **Association, not causation.** The model finds patterns in historical data. A high risk score does not show that a price, service or contract causes a customer to leave, so actions should be piloted with a control group.
- **Missing fields.** The dataset does not include satisfaction, complaints, support tickets, late payments, NPS, region or referral data, so those factors could not be tested. Tech Support, payment method and paperless billing were used as the nearest proxies.
- **In-sample scoring on the Retention Dashboard.** The dashboard scores all 7,043 customers, including the 80% used for training, so it shows 1,477 High or Critical customers and $76,429.55 of expected monthly revenue at risk. Out-of-sample, the test set has 296 High or Critical customers and about $15,030. Quote the held-out figures as the expected performance.
- **Single snapshot.** The data is one point in time, so the model cannot see how a customer's status changed over time.